

# Deep Learning Summer School 2026

## on AI for Economics and Finance

### Reading List

Torino, Italy · August 24–26, 2026

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#### Before the Summer School (Pre-Reading & Pre-Coding)

**Everyone should complete these two items before arriving in Torino.**

- **Coding — neural networks for economic models.** Simon Scheidegger, tutorials on using neural networks to solve simple Brock–Mirman models (Jupyter notebooks).  
Read and run both notebooks: Notebook 1 — [deterministic Brock–Mirman](#), and Notebook 2 — [stochastic Brock–Mirman](#). New to Python? Work through the [Python refresher](#) in this repository first, together with the [introduction to Jupyter notebooks](#).
- **Reading — mathematical background.** Marc Peter Deisenroth, A. Aldo Faisal & Cheng Soon Ong, *Mathematics for Machine Learning*, Cambridge University Press, 2020 (full text free online). [\[link\]](#)  
Skim **Chapters 2–5** (linear algebra, analytic geometry, matrix decompositions, vector calculus) and **Chapter 6** (probability). This is the level of fluency the lectures assume.

#### Recommended background

- Goodfellow, Bengio & Courville (2016). *Deep Learning*. MIT Press (full text free online). [\[link\]](#)  
The canonical deep-learning textbook. Skim Part I and Chapters 6–8 (feedforward networks, regularization, optimization/SGD). Chapter 10 (sequence modelling with recurrent nets) is useful preparation for Days 2 and 3.

#### Day 1 — Deep Equilibrium Nets, Deep Surrogates, and Heterogeneous Agents

*Simon Scheidegger (University of Lausanne), Yucheng Yang (University of Zurich)*

#### Introduction to Deep Equilibrium Nets, and hands-on exercises (morning)

- Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets*. International Economic Review 63(4), 1471–1525. Code: [github.com/sischei/DeepEquilibriumNets](https://github.com/sischei/DeepEquilibriumNets). [\[link\]](#)  
The flagship paper the DEQN lectures are built on: neural networks trained unsupervised to satisfy equilibrium conditions along simulated paths, with heterogeneity, uncertainty, and occasionally binding constraints.
- Scheidegger (2026). *Deep Learning for Solving and Estimating Dynamic Models in Economics and Finance*. arXiv:2605.14493. Code: [github.com/sischei/Deep\\_Learning\\_for\\_Solving\\_And\\_Estimating\\_Dynamic\\_Economic\\_Models](https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models). [\[link\]](#)

Lecture-note companion to the whole first half of the school; the hands-on exercises are drawn from this repository.

- Maliar, Maliar & Winant (2021). *Deep learning for solving dynamic economic models*. Journal of Monetary Economics 122, 76–101. [\[link\]](#)

### Deep Surrogates to Estimate Dynamic Models (afternoon)

- Chen, Didisheim & Scheidegger (2026). *Deep Surrogates for Finance: With an Application to Option Pricing*. Journal of Financial Economics 177, 104222. [\[link\]](#)

Neural-network surrogates approximate expensive structural models to accelerate estimation by orders of magnitude, here applied to a high-dimensional option pricing model.

- Chen, Cheng, Liu & Tang (2026). *Teaching Economics to the Machines*. NBER Working Paper No. 34713. [\[link\]](#)

A transfer-learning framework that injects the economic restrictions of a structural model into a machine-learning model: pre-train on synthetic data from the structural model, then fine-tune on empirical data.

### Heterogeneous agent models: DeepHAM and structural reinforcement learning (afternoon)

- Han, Yang & E (2026). *DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks*. Quantitative Economics 17(2), 297–341. Code: [github.com/frankhan91/DeepHAM](https://github.com/frankhan91/DeepHAM). [\[link\]](#)

Neural networks approximate value and policy functions and represent the cross-sectional distribution via learned generalized moments. Focus on the moment representation and the simulation-based training objective.

- Yang, Wang, Schaab & Moll (2025). *Structural Reinforcement Learning for Heterogeneous Agent Macroeconomics*. arXiv:2512.18892. [\[link\]](#)

A structural RL method that uses low-dimensional prices as state variables and lets agents learn equilibrium price dynamics from simulated paths (Krusell–Smith, Huggett with aggregate shocks, HANK).

### Recommended background

- Aiyagari (1994). *Uninsured Idiosyncratic Risk and Aggregate Saving*. Quarterly Journal of Economics 109(3), 659–684. [\[link\]](#)
- Krusell & Smith (1998). *Income and Wealth Heterogeneity in the Macroeconomy*. Journal of Political Economy 106(5), 867–896. [\[link\]](#)

The classic heterogeneous-agent model with aggregate risk; its curse of dimensionality is exactly what the deep-learning solvers are designed to overcome.

- Judd (1998). *Numerical Methods in Economics*. MIT Press. [\[link\]](#)
- Sutton & Barto (2018). *Reinforcement Learning: An Introduction*. 2nd ed., MIT Press (full text free online). [\[link\]](#)

## Day 2 — PDEs, Continuous Time, and the Foundations of Sequence Modelling

Simon Scheidegger (University of Lausanne), Yucheng Yang (University of Zurich), Markus Leippold (University of Zurich)

### Physics-informed neural networks for solving partial differential equations (morning)

- Raissi, Perdikaris & Karniadakis (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*. Journal of Computational Physics 378, 686–707. [\[link\]](#)

The paper that started the PINN literature. Read Sections 1–3 carefully; the residual-based loss it introduces is the workhorse of the entire continuous-time toolkit.

### Deep learning for continuous-time models (late morning)

- Achdou, Han, Lasry, Lions & Moll (2022). *Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach*. Review of Economic Studies 89(1), 45–86. [\[link\]](#)

Recasts the Aiyagari–Bewley–Huggett model in continuous time as coupled HJB and Kolmogorov-forward PDEs, the continuous-time heterogeneous-agent toolkit underlying this session.

- Gu, Laurière, Merkel & Payne (2024). *Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models*. arXiv:2406.13726. [\[link\]](#)

Introduces Economic-Model-Informed Neural Networks (EMINNs) to compute global solutions of master equations, with the continuous-time Krusell–Smith model as the lead application.

- Payne, Rebei & Yang (2025). *Deep Learning for Search and Matching Models*. Swiss Finance Institute Research Paper No. 25-05. [\[link\]](#)

Extends the deep-learning-for-PDE approach to labor-market models with aggregate shocks and two-sided heterogeneity, and estimates them by simulated method of moments.

### Foundations of Sequence Modelling: From RNNs to Transformers (afternoon)

- Hochreiter & Schmidhuber (1997). *Long Short-Term Memory*. Neural Computation 9(8), 1735–1780. [\[link\]](#)

The gated recurrent architecture that made long-range dependence learnable, and the baseline against which attention is motivated.

- Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser & Polosukhin (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems 30. [\[link\]](#)

The Transformer. Everything on Day 3 is a variation on this architecture, so read it before the LLM sessions.

*Further readings for this session will be added by the lecturer before the school.*

## Day 3 — Large Language Models and Deep Sequence Modelling in Finance

Markus Leippold (University of Zurich), Fabio Trojani (University of Geneva)

### The LLM Revolution: Scaling Laws and Modern Architectures (morning)

- Devlin, Chang, Lee & Toutanova (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Proceedings of NAACL-HLT 2019, 4171–4186. [\[link\]](#)
- Brown, Mann, Ryder et al. (2020). *Language Models are Few-Shot Learners*. Advances in Neural Information Processing Systems 33. [\[link\]](#)

The GPT-3 paper: the counterpart to BERT’s encoder paradigm, and the origin of in-context learning.

- Kaplan, McCandlish, Henighan et al. (2020). *Scaling Laws for Neural Language Models*. arXiv:2001.08361. [\[link\]](#)

Read alongside [Hoffmann et al. \(2022\)](#), “Training Compute-Optimal Large Language Models” (Chinchilla), which revises the compute-optimal trade-off between model size and data.

*Further readings for this session will be added by the lecturer before the school.*

### Domain Adaptation: Building Financial & Climate Intelligence (late morning)

- Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang & Chen (2022). *LoRA: Low-Rank Adaptation of Large Language Models*. International Conference on Learning Representations (ICLR). [\[link\]](#)

The parameter-efficient fine-tuning method used in the hands-on part of this session.

- Webersinke, Kraus, Bingler & Leippold (2022). *ClimateBert: A Pretrained Language Model for Climate-Related Text*. Proceedings of the AAAI 2022 Fall Symposium, arXiv:2110.12010. Models: [huggingface.co/climatebert](https://huggingface.co/climatebert). [\[link\]](#)

The instructor’s own domain-adaptation case study: further pretraining on two million climate-related paragraphs, and what it buys on downstream classification tasks.

- Ouyang, Wu, Jiang et al. (2022). *Training language models to follow instructions with human feedback*. Advances in Neural Information Processing Systems 35. [\[link\]](#)

The InstructGPT/RLHF paper. Pair it with [Rafailov et al. \(2023\)](#), “Direct Preference Optimization”, the simpler alignment objective that has largely replaced it in practice.

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### Advanced Frontiers: From RAG to Reasoning and Agentic LLMs (afternoon)

- Lewis, Perez, Piktus et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing Systems 33. [\[link\]](#)
- Vaghefi, Stambach, Muccione et al. (2023). *ChatClimate: Grounding conversational AI in climate science*. Communications Earth & Environment 4, 480. [\[link\]](#)

RAG applied end to end in a domain setting: GPT-4 grounded in the IPCC AR6 report, with the answers evaluated by IPCC authors. The template for the applied part of this session.

- Yao, Zhao, Yu, Du, Shafran, Narasimhan & Cao (2023). *ReAct: Synergizing Reasoning and Acting in Language Models*. International Conference on Learning Representations (ICLR). [\[link\]](#)

Interleaved reasoning traces and tool calls, the basic loop behind agentic LLMs. Read after [Wei et al. \(2022\)](#), “Chain-of-Thought Prompting”.

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### Using Deep Sequence Modelling in Finance (afternoon)

- Gaegauf, Scheidegger & Trojani (2023). *A Comprehensive Machine Learning Framework for Dynamic Portfolio Choice With Transaction Costs*. Swiss Finance Institute Research Paper No. 23-114. [\[link\]](#)

The instructor’s own framework: Gaussian-process regression plus Bayesian active learning inside a dynamic programming algorithm, which characterizes the irregular no-trade region and scales to many risky assets.

- Gu, Kelly & Xiu (2020). *Empirical Asset Pricing via Machine Learning*. Review of Financial Studies 33(5), 2223–2273. [\[link\]](#)

The benchmark comparison of machine-learning methods for return prediction, and the reference point for what deep sequence models must beat.

- Leippold (2023). *Sentiment spin: Attacking financial sentiment with GPT-3*. Finance Research Letters 55, Part B. [\[link\]](#)

A cautionary counterpart to the LLM sessions: dictionary-based financial sentiment measures can be adversarially manipulated with close to 99% success, while context-aware models remain robust.

*Further readings for this session will be added by the lecturer before the school.*

### Recommended background

- Gentzkow, Kelly & Taddy (2019). *Text as Data*. Journal of Economic Literature 57(3), 535–574. [\[link\]](#)

The standard survey of how economists turn text into data, and the natural bridge from Day 3’s methods to empirical work in economics and finance.

- Chen, Pelger & Zhu (2024). *Deep Learning in Asset Pricing*. Management Science 70(2), 714–750. [\[link\]](#)